

JAMSHEDPUR, India

WMO: 427980

Lat: 22.813N

Lon: 86.169E

Elev: 505

StdP: 14.43

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.1	52.2	35.0	30.3	82.7	38.4	34.8	80.2	7.8	73.3	7.0	73.6	0.2	270	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	21.9	107.8	72.7	105.1	73.3	102.3	74.2	82.8	92.4	82.0	90.9	81.4	89.8	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	162.9	86.7	80.0	158.6	85.7	79.3	155.1	84.9	47.6	92.6	46.6	91.3	45.8	90.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	7.9	6.9	46.4	111.8	1.1	2.2	45.6	113.5	45.0	114.8	44.3	116.0	43.5	117.6		
(5)			44.5	85.0	1.4	0.7	43.5	85.5	42.7	86.0	41.9	86.4	40.9	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.3	66.1	73.1	81.7	89.4	90.4	88.5	84.3	83.9	83.6	80.7	74.0	67.1		
	DBStd	8.79	3.45	4.54	4.45	4.42	4.13	5.00	2.53	2.30	2.51	3.10	3.67	3.32		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	45	27	2	0	0	0	0	0	0	0	0	0	16		
	CDD50	11042	499	647	983	1181	1252	1154	1063	1050	1009	952	721	531		
	CDD65	5613	62	229	518	731	787	704	598	585	559	487	271	82		
	CDH74	67600	577	2170	6159	10492	11345	9673	7053	6747	6274	4644	1875	591		
	CDH80	33755	95	893	3433	6656	7114	5500	2805	2518	2380	1743	545	73		
(14) Wind	WSAvg	1.8	1.0	1.3	1.8	2.6	2.8	2.7	2.4	2.1	1.9	1.1	0.7	0.8		
(15) Precipitation	PrecAvg	57.0	0.7	0.6	0.7	1.0	3.3	8.7	13.8	12.7	11.1	3.2	0.7	0.4		
	PrecMax	79.5	5.2	3.4	3.9	4.9	6.9	25.2	29.4	23.4	19.2	11.5	5.4	3.8		
	PrecMin	25.3	0.0	0.0	0.0	0.0	0.0	1.3	3.8	5.1	4.4	0.6	0.2	0.0		
	PrecStd	12.1	1.1	0.8	1.0	1.3	1.9	5.3	5.8	4.7	4.1	2.8	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	96.4	104.3	111.4	110.2	110.2	96.5	94.5	95.0	94.0	89.8	84.6		
		MCWB	64.0	66.6	69.4	69.9	72.3	75.5	81.1	80.7	79.0	75.4	71.6	64.8		
	2%	DB	82.7	92.0	100.4	107.9	107.0	106.0	93.6	92.4	92.9	91.7	87.3	82.0		
		MCWB	62.9	66.0	68.7	70.3	73.6	77.9	80.8	80.7	79.5	76.3	70.3	63.9		
	5%	DB	79.8	88.8	97.3	105.3	104.4	101.5	91.5	90.7	91.2	90.0	85.4	79.8		
		MCWB	61.4	65.0	68.0	70.6	75.0	79.4	80.5	80.6	79.6	76.1	69.6	63.0		
	10%	DB	77.2	85.5	94.7	102.4	101.4	97.8	89.6	89.1	89.6	88.2	83.4	77.5		
		MCWB	60.6	64.4	67.7	71.0	76.6	79.8	80.3	80.3	79.5	75.4	68.7	62.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.2	73.5	79.1	81.2	83.9	84.5	83.0	82.7	82.3	80.7	75.5	69.9		
		MCDB	77.4	84.1	91.0	91.4	94.6	95.0	91.3	89.9	90.0	88.1	83.3	78.5		
	2%	WB	66.4	70.1	75.4	79.8	82.5	83.1	81.9	81.8	81.2	79.6	74.1	67.5		
		MCDB	75.1	81.3	87.6	90.3	93.0	93.7	89.3	88.8	88.4	86.6	81.6	75.7		
	5%	WB	64.7	68.3	73.3	78.5	81.4	82.0	81.2	81.1	80.7	78.9	72.8	65.7		
		MCDB	74.3	80.1	86.2	89.6	92.5	92.2	88.2	87.7	87.7	85.5	80.2	74.8		
	10%	WB	63.1	66.7	71.4	77.0	80.1	81.1	80.7	80.7	80.1	78.1	71.6	64.1		
		MCDB	73.5	79.7	85.2	89.6	92.2	91.2	87.2	87.0	86.6	84.3	79.1	73.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.9	25.0	26.1	25.6	21.9	15.7	10.3	10.2	11.5	15.6	21.3	24.1		
		MCDBR	28.3	29.0	29.3	28.9	25.4	22.1	13.5	12.9	14.3	17.6	21.9	25.8		
	5% WB	MCWBR	10.7	8.9	8.5	8.4	7.8	6.5	4.2	3.9	4.2	5.4	7.3	9.5		
		MCWBR	22.2	22.9	24.6	25.1	21.7	17.7	11.4	11.2	12.3	13.7	17.1	21.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.527	0.513	0.560	0.593	0.651	0.679	0.593	0.579	0.587	0.589	0.599	0.571		
		taud	1.845	1.917	1.839	1.814	1.739	1.707	1.905	1.955	1.917	1.853	1.748	1.778		
	Ebn at Noon	Ebn at Noon	229	246	241	235	220	213	232	236	231	222	207	210		
		Edh at Noon	60	59	67	69	74	76	62	59	61	62	65	62		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1265	1548	1832	2035	2014	1614	1311	1349	1390	1436	1305	1179		
	RadStd	RadStd	84	91	78	86	93	134	109	74	88	130	70	76		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	+0.59	+0.68	N/A	N/A	N/A	N/A

Nomenclature: See separate page